COURSE SLIDES · WSQ

R Fundamental and Statistical Analysis for Beginners

Course Code: TGS-2019504058

Trainer: Dr Alfred Ang

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Tertiary Infotech Academy Pte Ltd

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Digital Attendance (Mandatory)

1

AM attendance

Scan the SSG digital attendance QR code at the start of each day.

2

PM attendance

Scan again after lunch when instructed by the trainer.

3

Assessment attendance

Complete the separate assessment digital attendance before sitting the papers.

4

Requirement

At least 75% attendance and a Competent result are required for funding outcomes.

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About the Trainer



Your Trainer

Complete before class

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

AREAS OF EXPERTISE _____

TRAINING & INDUSTRY EXPERIENCE _____

CONTACT _____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Data Science & AI

NAME	Dr Alfred Ang
TITLE / DESIGNATION	Principal Trainer
QUALIFICATIONS	Ph.D. in Operations Research · ACTA/ACLP training experience
AREAS OF EXPERTISE	R, statistics, analytics, AI and optimisation
TRAINING & INDUSTRY EXPERIENCE	University, professional training and analytics consultancy
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LEARNER INTRODUCTION

Let's Know Each Other

1

You

Name, role and organisation.

2

Experience

Your prior exposure to data, spreadsheets or code.

3

Question

One statistical question you would like R to answer.

4

Outcome

One practical capability you want by Day 2.

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HOUSEKEEPING

Ground Rules

1

Be present

Return from breaks on time and meet the attendance requirement.

2

Participate

Try the code, ask questions and share evidence.

3

Respect

One conversation at a time; disagree with ideas, not people.

4

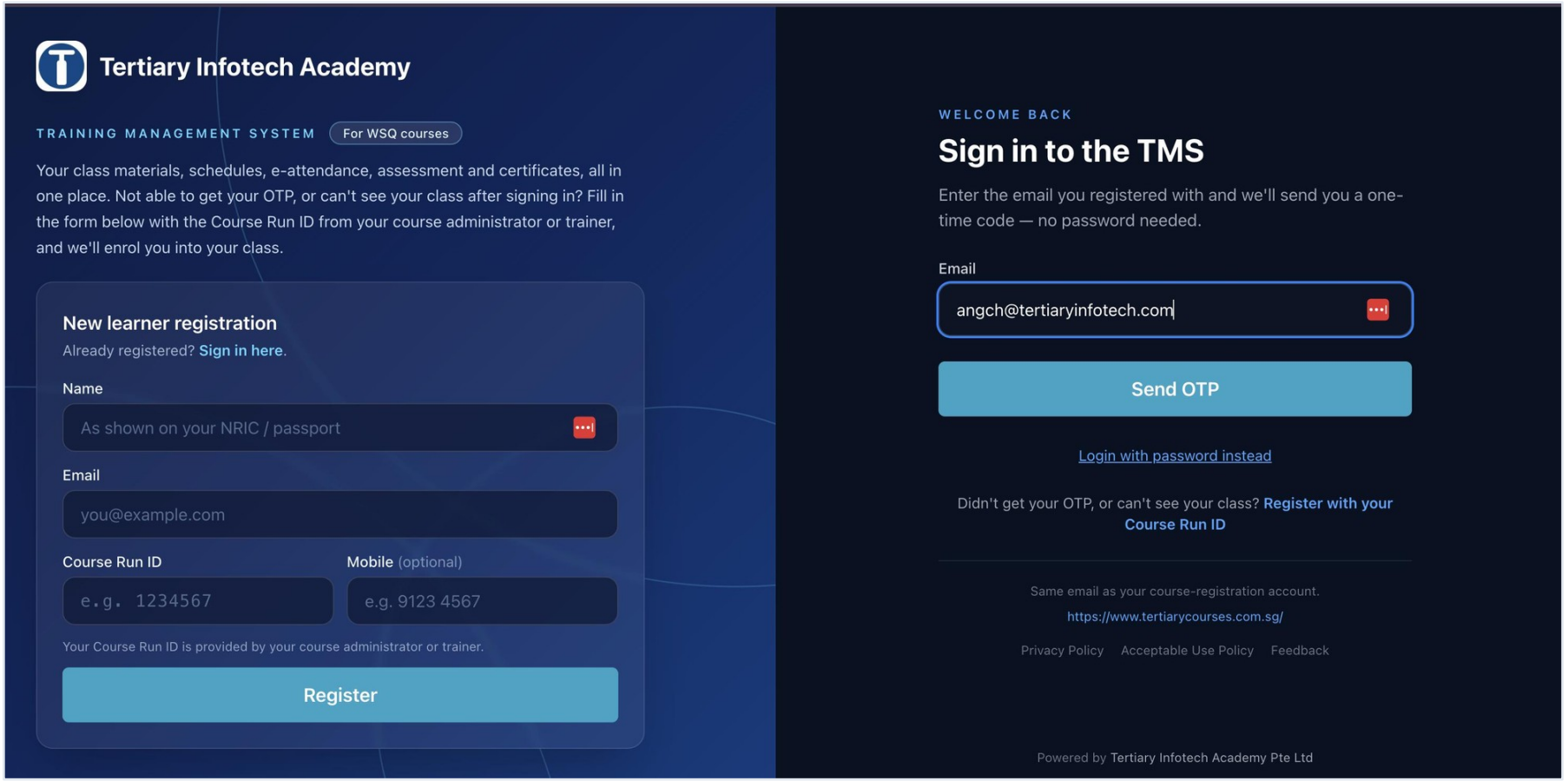
Protect focus

Set devices to silent and step out quietly when necessary.

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COURSE PORTAL

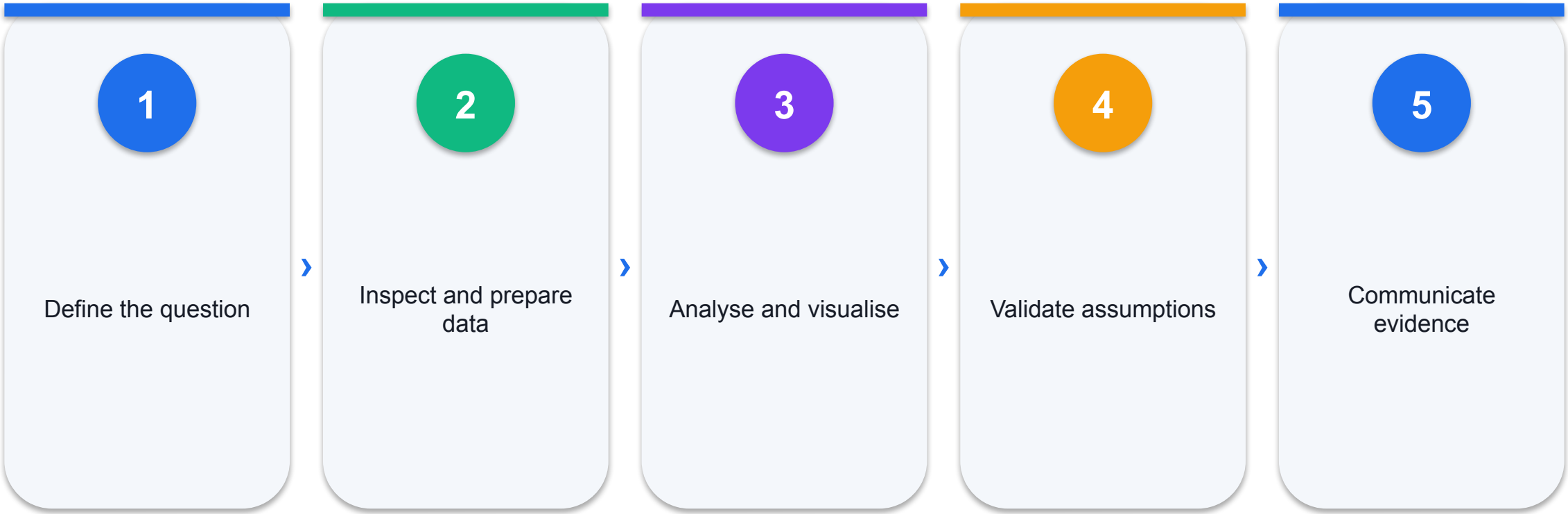
Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Analysis Workflow



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WHAT YOU WILL ACHIEVE

Learning Outcomes

1

LO1

Install R and RStudio IDE and work confidently with scripts, the console and help system.

2

LO2

Select, create, inspect and manipulate R data types and data structures.

3

LO3

Use R packages and datasets and import, validate and export external data.

4

LO4

Create and interpret appropriate data visualisations in R.

5

LO5

Write R programs using conditions, loops and reusable functions.

6

LO6

Apply descriptive statistics, correlation, regression, hypothesis testing and ANOVA in R.

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16 HOURS · 2 DAYS

Lesson Plan — 2 Days

DAY 1 · FOUNDATIONS

09:00–18:00
Topics 1–4: environment, data structures, packages/data I/O and visualisation.
Labs 1–8 with breaks and lunch.

DAY 2 · PROGRAMMING & STATISTICS

09:00–15:40 teaching and Labs 9–12.
15:40–18:00 final assessment: WA 60 min + Case Study 80 min.

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BEFORE THE ASSESSMENT

Briefing for Assessment

1

Individual

No discussion, photos or recording of assessment scripts.

2

Open book

Use the slides, Learner Guide and approved materials only.

3

Submission

Complete the provided document and upload it to the LMS.

4

Conduct

Place phones away; use black/blue pen for hard copy; no correction fluid.

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OPEN BOOK

Assessment and Funding

1

Written Assessment

7 open-ended SAQs · K1–K7 · 60 minutes.

2

Case Study

6 open-ended tasks · A1–A10 · 80 minutes.

3

Assessment result

Competent (C) or Not Yet Competent (NYC); reassessment and appeal pathways apply.

4

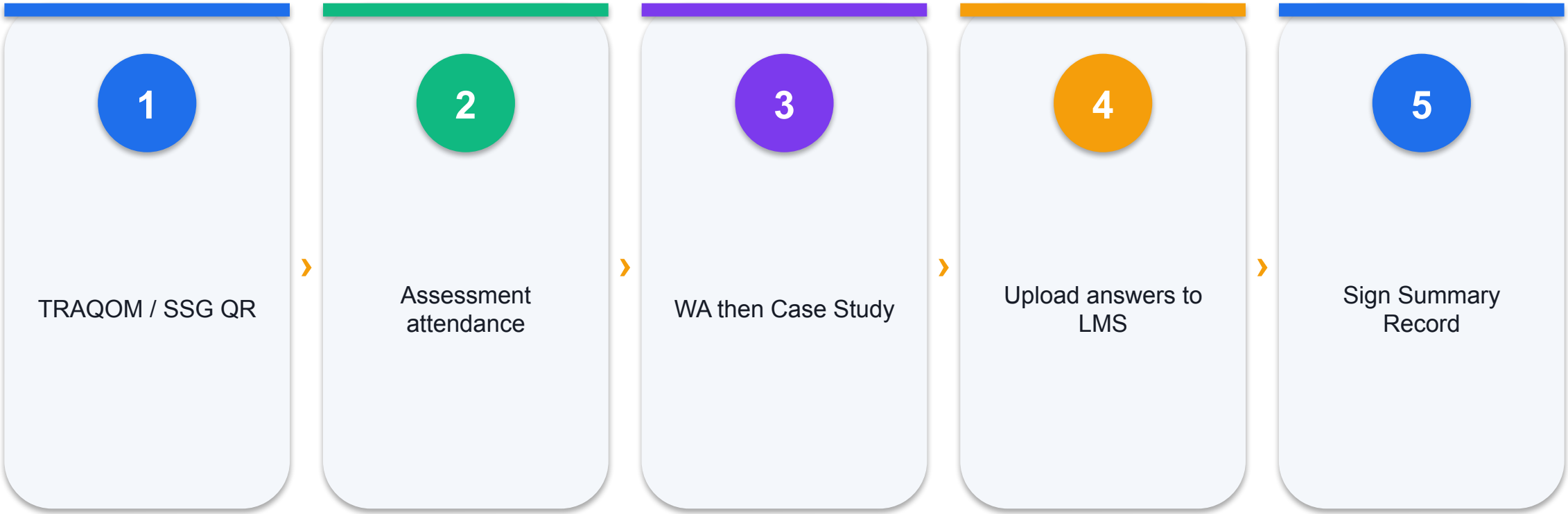
Funding condition

At least 75% digital attendance and a Competent result.

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ASSESSMENT DAY

Assessment Flow



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Access the Hands-On Labs

OPTION A · CLONE

git clone https://github.com/tertiarycourses/TGS-2019504058-R-Fundamental-and-Statistical-Analysis-for-Beginners.git

Open the labs/ folder in RStudio.

Open the GitHub lab repository →

OPTION B · DOWNLOAD ZIP

Open the GitHub repository, choose Code → Download ZIP, extract it, then open the lab folder.

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SOURCE-ALIGNED REBUILD

Course Source and Design Boundary

1

Legacy coverage

The v14 deck and original exercise pack are the approved instructional baseline.

2

Current outline

The live course page confirms 2 days, 16 hours, six learning outcomes and six topics.

3

Current design

Every slide is rebuilt with editable visual components; the legacy theme is not copied.

4

Procedure boundary

Detailed steps live in the Learner Guide and lab folders; slides teach concepts and decisions.

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TOPIC 01 · K4 · A2

Getting Started in R

R · RStudio · scripts · comments · help

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What is R?

1

Meaning

An open-source language and environment for statistical computing, graphics and reproducible analysis.

2

How it works

Objects live in memory; functions transform them; scripts preserve the exact sequence of work.

3

Decision rule

Use R when the task combines data preparation, statistical reasoning, visualisation and repeatable reporting.

4

Watch out

R is case-sensitive: mean(x) and Mean(x) are different names.

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Apply: What is R?

Interpretation

A script turns an interactive calculation into an auditable analysis.

R PATTERN

```
x <- c(12, 15, 18)
mean(x)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Install R and RStudio

1

Meaning

R is the computation engine; RStudio is an integrated development environment that makes the engine easier to use.

2

How it works

Install R first, then RStudio Desktop. RStudio detects the installed R version.

3

Decision rule

Confirm both versions and keep packages in a user library that you can update.

4

Watch out

Installing only RStudio does not install the R engine.

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Apply: Install R and RStudio

Interpretation

Verify the engine before troubleshooting packages or scripts.

R PATTERN

```
R.version.string
R.home()
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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RStudio Interface

1

Meaning

The Source, Console, Environment/History and Files/Plots/Packages/Help panes separate authoring, execution, objects and output.

2

How it works

Write durable code in Source; execute selected lines; inspect objects and plots without mixing them into the script.

3

Decision rule

Save the project and script before running a new analysis.

4

Watch out

Commands typed only in Console are easy to lose and difficult to audit.

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Apply: RStudio Interface

Interpretation

Use the Source pane as the record of analysis; use the Console as feedback.

R PATTERN

```
getwd()
ls()
objects()
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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R Scripts and Comments

1

Meaning

An .R script is a plain-text program containing valid R expressions and comments.

2

How it works

Run one line, a selection or the whole script. Use # to explain intent and assumptions.

3

Decision rule

Organise scripts into setup, import, validation, analysis, visualisation and export sections.

4

Watch out

Comments should explain why a choice was made, not merely repeat the code.

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Apply: R Scripts and Comments

Interpretation

Readable scripts are easier to review, rerun and repair.

R PATTERN

```
# Average valid measurements
avg <- mean(x, na.rm = TRUE)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Help and Self-Diagnosis

1

Meaning

R includes searchable documentation, examples and object inspection tools.

2

How it works

Use help(), ?, args(), example(), str() and traceback() to move from symptom to evidence.

3

Decision rule

Inspect the function signature and the object structure before changing code.

4

Watch out

Copying an online snippet without checking the installed package version can create silent mismatches.

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Apply: Help and Self-Diagnosis

Interpretation

Good R users do not memorise everything; they diagnose efficiently.

R PATTERN

```
help("plot")
args(mean)
str(mtcars)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Lab 1: Your First Reproducible R Script

1

Scenario

A training coordinator needs a repeatable summary of course attendance.

2

Output

A saved .R script that creates, summarises and exports attendance data.

3

Concepts

objects, assignment, Source vs Console, comments, help

4

Mapping

LO1 · K4 · A2

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VERIFY THE RESULT

Lab 1: Evidence and Acceptance

1

Evidence 1

Script runs from a clean R session without manual Console steps.

2

Evidence 2

Summary excludes the missing value and reports the expected valid count.

3

Evidence 3

Evidence sheet records the output filename and verification result.

4

Detailed procedure

Follow the Learner Guide and labs/lab-01-rstudio-first-script/README.pdf. No click-by-click steps are placed in these slides.

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Lab 2: Help, Objects and Session Diagnostics

1

Scenario

A colleague's script fails because an object and function call are misunderstood.

2

Output

A diagnostic R script and evidence log explaining the root cause.

3

Concepts

help, args, str, class, sessionInfo

4

Mapping

LO1 · K4 · A2

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VERIFY THE RESULT

Lab 2: Evidence and Acceptance

1

Evidence 1

Diagnostics identify the missing argument or wrong object type.

2

Evidence 2

Corrected expression returns the expected value.

3

Evidence 3

Session information is captured for reproducibility.

4

Detailed procedure

Follow the Learner Guide and labs/lab-02-help-session-diagnostics/README.pdf. No click-by-click steps are placed in these slides.

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Topic 01 Recap

1

Explain

Describe the central ideas in Getting Started in R using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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02

TOPIC 02 · K3 · K6 · K7 · A5 · A9

Data Types

atomic values · vectors · matrices · arrays · data frames · lists · factors

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Atomic Values and Variables

1

Meaning

R stores logical, integer, double, complex, character and raw atomic values.

2

How it works

Assignment binds a name to a value; typeof(), class() and str() expose different aspects of the object.

3

Decision rule

Use descriptive snake_case names and explicit conversion at system boundaries.

4

Watch out

Coercion can change the entire vector when one incompatible value is added.

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Apply: Atomic Values and Variables

Interpretation

Type is a property of the object, not the variable name.

R PATTERN

```
count <- 12L
rate <- 0.35
label <- "May"
typeof(rate)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Character Data

1

Meaning

Character vectors hold text; quotes delimit values and escaping represents special characters.

2

How it works

paste(), paste0(), nchar(), substr(), strsplit() and gsub() create, inspect and transform strings.

3

Decision rule

Use factors for repeated categorical levels; keep free text as character.

4

Watch out

A missing value NA is not the same as the literal string "NA".

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Apply: Character Data

Interpretation

Text cleaning should preserve meaning while making categories consistent.

R PATTERN

```
email <- "learner@example.com"
parts <- strsplit(email, "@")[[1]]
paste(parts, collapse = " at ")
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Vectors

1

Meaning

A vector is a one-dimensional, homogeneous sequence and the foundation of R computation.

2

How it works

c(), :, seq() and rep() create vectors; vectorised operators act element by element.

3

Decision rule

Prefer vectorised operations for clarity and performance when every element follows the same rule.

4

Watch out

Recycling a shorter vector may produce a result that runs but is conceptually wrong.

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Apply: Vectors

Interpretation

Think in complete columns rather than one cell at a time.

R PATTERN

```
scores <- c(72, 84, 91, 67)
scaled <- scores / 100
mean(scaled)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Indexing and Missing Values

1

Meaning

Square brackets select by position, name or logical condition; negative indices exclude.

2

How it works

is.na() identifies missing values and na.rm = TRUE tells many summaries to omit them.

3

Decision rule

Make the missing-data rule explicit before calculating a statistic.

4

Watch out

Comparing x == NA never identifies missing values; use is.na(x).

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Apply: Indexing and Missing Values

Interpretation

Filtering is part of the analysis logic and must be documented.

R PATTERN

```
x <- c(4, NA, 9, -2, 7)
x[!is.na(x) & x > 0]
mean(x, na.rm = TRUE)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Vector Statistics and Arithmetic

1

Meaning

Summary functions collapse a vector; cumulative functions preserve a running path.

2

How it works

mean, median, sum, min, max, quantile, sort, diff, cumsum and rank answer different questions.

3

Decision rule

Choose robust summaries such as median and IQR when outliers distort the mean.

4

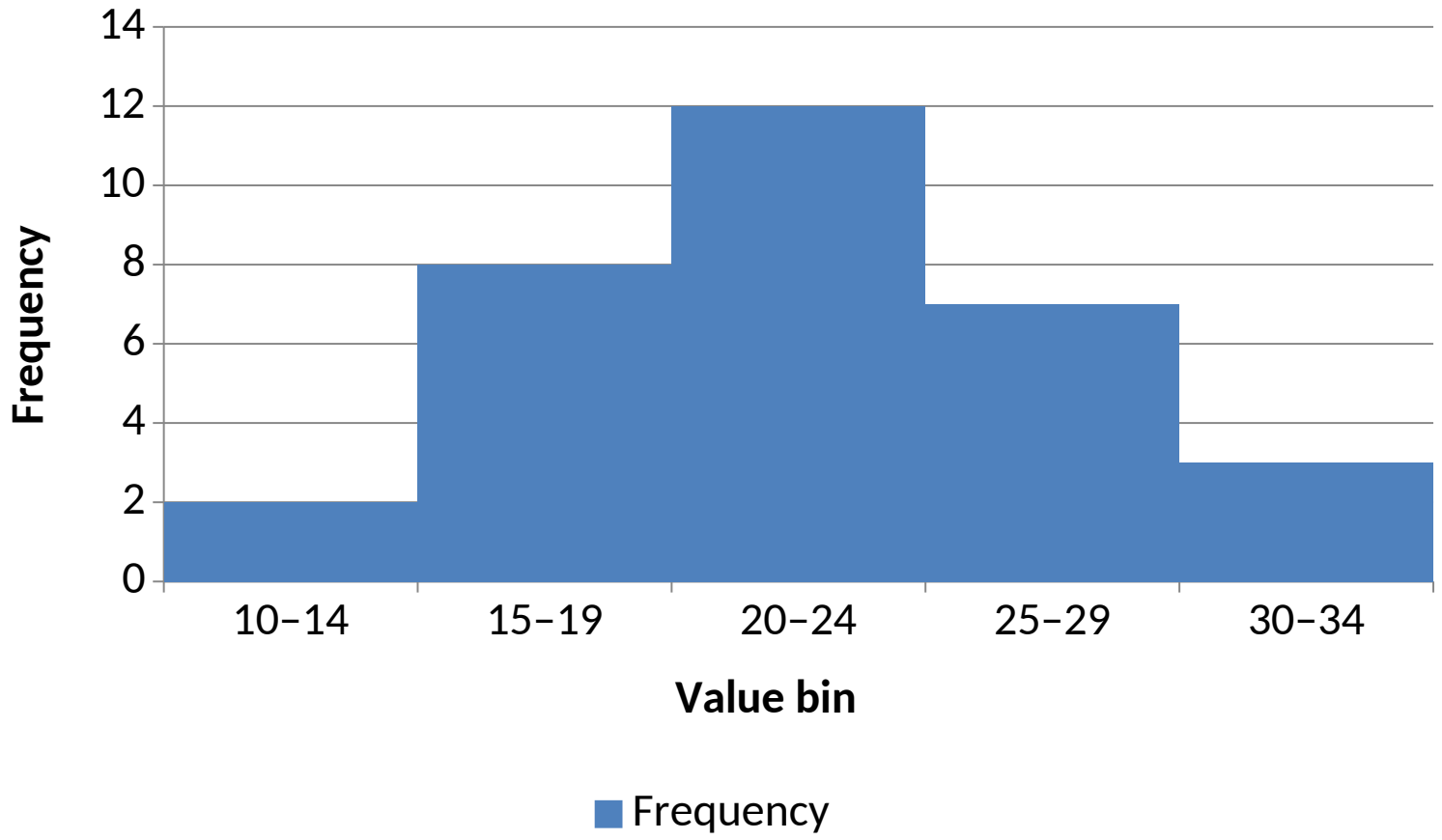
Watch out

A plausible number can still answer the wrong statistical question.

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EDITABLE POWERPOINT CHART

Apply: Vector Statistics and Arithmetic



READ THE EVIDENCE

Compare location and spread before making a claim about typical values.

Native chart · editable data

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TOPIC 02 · K3 · K6 · K7 · A5 · A9

Matrices

1

Meaning

A matrix is a two-dimensional homogeneous object with rows and columns.

2

How it works

matrix(), rbind() and cbind() construct matrices; [row, column] selects elements.

3

Decision rule

Use matrices for numeric linear algebra; use data frames for mixed-type records.

4

Watch out

R fills a matrix by column unless byrow = TRUE is specified.

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Apply: Matrices

Interpretation

Always state the dimension and fill direction when constructing a matrix.

R PATTERN

```
m <- matrix(1:12, nrow = 3, byrow = TRUE)
m[2, ]
m[, 3]
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Arrays

1

Meaning

An array generalises a matrix to three or more dimensions while keeping one atomic type.

2

How it works

dim specifies the length of each axis; commas separate the indices for each dimension.

3

Decision rule

Use an array for regularly shaped repeated measurements such as site × metric × month.

4

Watch out

Without dimension names, slices are difficult to interpret and easy to transpose.

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Apply: Arrays

Interpretation

Name dimensions whenever an axis carries business meaning.

R PATTERN

```
a <- array(1:18, dim = c(3, 2, 3))
a[, , 2]
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Data Frames

1

Meaning

A data frame is a two-dimensional table whose columns may have different types but equal length.

2

How it works

$`, [[ ]] and [rows, columns] access data; head(), tail(), names(), str() and summary() reveal structure.

3

Decision rule

Use a data frame for observational records such as one row per person or day.

4

Watch out

Row names are labels, not a reliable database key.

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Apply: Data Frames

Interpretation

Inspect column names and types before filtering or modelling.

R PATTERN

```
cars <- mtcars
small <- cars[c("mpg", "am", "wt")]
head(small)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Filtering and Subsetting

1

Meaning

Logical expressions select rows that meet one or more conditions.

2

How it works

Use &, | and ! element-wise; subset() can improve readability for simple interactive tasks.

3

Decision rule

Store complex conditions in named variables so the rule can be reviewed.

4

Watch out

&& and || examine only the first element and are normally wrong for row filtering.

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Apply: Filtering and Subsetting

Interpretation

A filter is a business rule expressed as code.

R PATTERN

```
keep <- mtcars$mpg > 15 & mtcars$am == 1
result <- mtcars[keep, c("mpg", "am", "wt")]
summary(result)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Lists

1

Meaning

A list can hold objects of different types, sizes and shapes.

2

How it works

list() creates; [[ ]] extracts one element; [ ] returns a sub-list; $ accesses a named element.

3

Decision rule

Use lists for model results, nested configuration and heterogeneous outputs.

4

Watch out

Confusing [ ] with [[ ]] often leaves a list where a vector or model was expected.

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Apply: Lists

Interpretation

Lists let one analysis result carry data, model and diagnostics together.

R PATTERN

```
result <- list(data = mtcars, model = lm(mpg ~
wt, mtcars))
coef(result$model)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Factors

1

Meaning

A factor represents categorical data as a fixed set of levels.

2

How it works

factor() defines levels and order; table() counts; relevel() changes the reference category.

3

Decision rule

Use factors when a category will drive grouping, plotting or a statistical model.

4

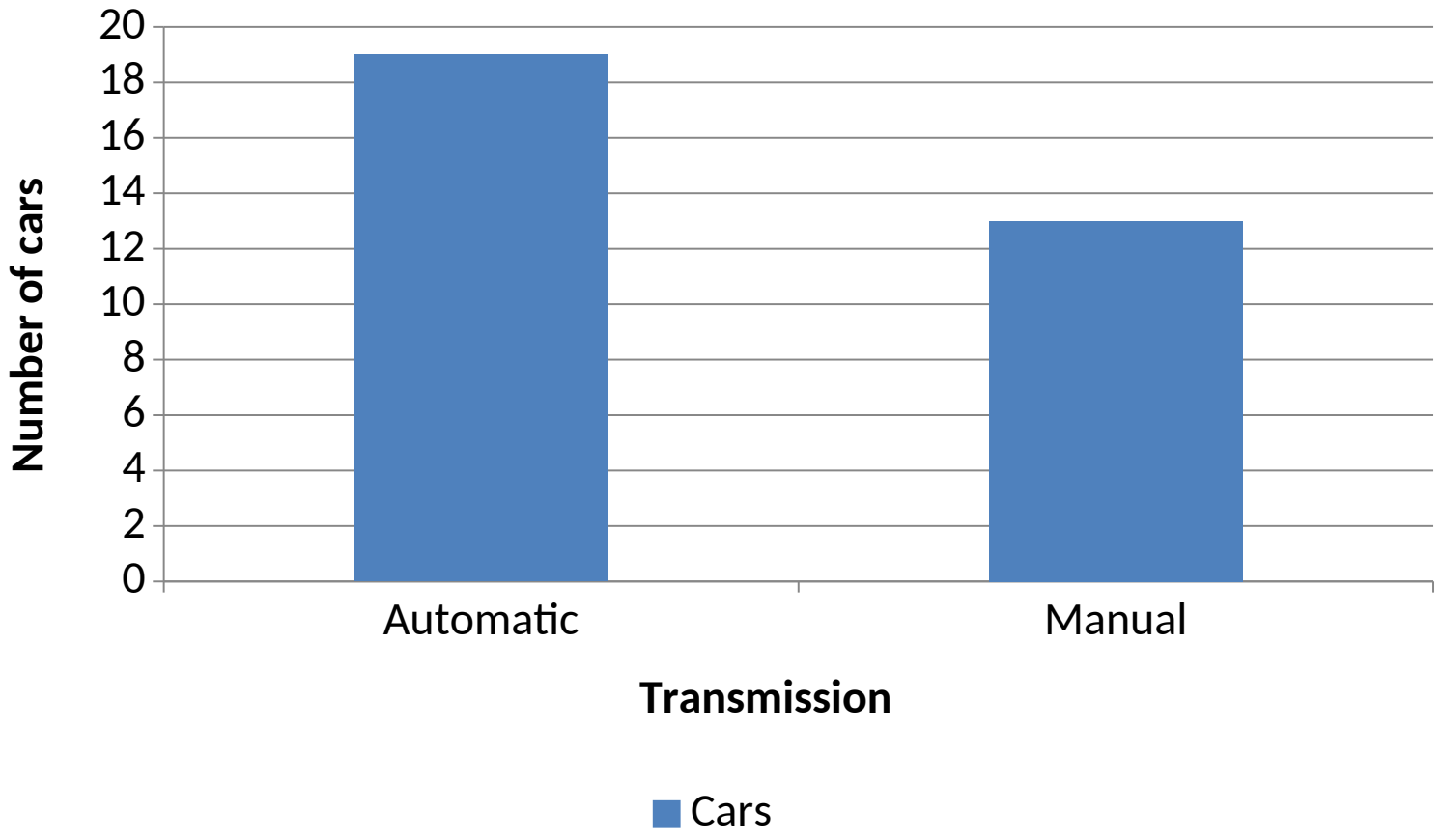
Watch out

Unused or misspelled levels can change plots and model coefficients.

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EDITABLE POWERPOINT CHART

Apply: Factors



READ THE EVIDENCE

Factor levels encode meaning, not just display order.

Native chart · editable data

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Choosing the Right Structure

1

Meaning

Shape, type consistency and access pattern determine the correct R structure.

2

How it works

Vector: 1D homogeneous; matrix/array: regular homogeneous dimensions; data frame: tabular mixed types; list: heterogeneous nesting.

3

Decision rule

Select the simplest structure that faithfully represents the data and intended operation.

4

Watch out

Forcing mixed data into a matrix coerces every value to a common type.

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Apply: Choosing the Right Structure

Interpretation

Data structure is an analytical design decision.

R PATTERN

```
objects <- list(vector=1:3,
matrix=matrix(1:4,2), frame=mtcars[1:2,1:3])
lapply(objects, class)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Lab 3: Vectors, Indexing and Missing Values

1

Scenario

An operations analyst must clean sensor readings before reporting an average.

2

Output

A clean vector, filtered subset and robust summary table.

3

Concepts

vectors, logical indexing, NA, mean, median, IQR

4

Mapping

LO2 · K3 · A5

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VERIFY THE RESULT

Lab 3: Evidence and Acceptance

1

Evidence 1

No invalid or missing reading enters the final mean.

2

Evidence 2

Mean, median and IQR match the expected values.

3

Evidence 3

All filtering rules are visible in the R script.

4

Detailed procedure

Follow the Learner Guide and labs/lab-03-vectors-indexing-missing/README.pdf. No click-by-click steps are placed in these slides.

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Lab 4: Data Structures and Factors

1

Scenario

A fleet analyst needs the correct R structure for mixed vehicle records.

2

Output

A typed data frame, a matrix slice and a factor frequency table.

3

Concepts

matrix, data frame, list, factor, subsetting

4

Mapping

LO2 · K3 · A5 · A9

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VERIFY THE RESULT

Lab 4: Evidence and Acceptance

1

Evidence 1

Matrix contains only numeric values and has the expected dimensions.

2

Evidence 2

Data frame preserves column types.

3

Evidence 3

Factor levels and counts are explicit and correct.

4

Detailed procedure

Follow the Learner Guide and labs/lab-04-structures-factors/README.pdf. No click-by-click steps are placed in these slides.

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K3 · K6 · K7 · A5 · A9 · CHECKPOINT

Topic 02 Recap

1

Explain

Describe the central ideas in Data Types using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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03

TOPIC 03 · K3 · K6 · A2 · A5 · A9

R Packages and Data I/O

packages · built-in datasets · working directories · CSV/Excel · validation · export

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Packages and Libraries

1

Meaning

A package bundles functions, data and documentation; a library is the installed location and attached package set.

2

How it works

install.packages() installs once; library() attaches per session; package::function calls explicitly.

3

Decision rule

Prefer explicit package::function in shared scripts when function names may collide.

4

Watch out

Running install.packages() on every execution slows work and can change the environment unexpectedly.

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Apply: Packages and Libraries

Interpretation

Separate environment setup from analysis execution.

R PATTERN

```
if (!requireNamespace("readxl", quietly=TRUE))
install.packages("readxl")
library(readxl)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Built-in Datasets

1

Meaning

R ships documented datasets such as mtcars, quakes, sleep and chickwts for reproducible examples.

2

How it works

data() lists available data; ?mtcars explains variables; str() validates structure.

3

Decision rule

Read the dataset documentation before interpreting a column or unit.

4

Watch out

Familiarity with a dataset name does not replace checking its definitions.

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Apply: Built-in Datasets

Interpretation

Metadata is part of the dataset.

R PATTERN

```
data(mtcars)
?mtcars
str(mtcars)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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TOPIC 03 · K3 · K6 · A2 · A5 · A9

Working Directory and Paths

1

Meaning

The working directory is the base location used by relative file paths.

2

How it works

getwd() reports it; RStudio Projects make it stable; file.path() builds portable paths.

3

Decision rule

Prefer project-relative paths and avoid setwd() inside reusable scripts.

4

Watch out

Absolute user-specific paths break when another learner runs the script.

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Apply: Working Directory and Paths

Interpretation

Portable paths are essential for reproducible course labs.

R PATTERN

```
input_path <- file.path("data", "weather.csv")
file.exists(input_path)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Import and Validate Data

1

Meaning

Import converts external bytes into an R object; validation confirms that the object matches the analysis contract.

2

How it works

Read, inspect names/types/dimensions, check missingness and ranges, then analyse.

3

Decision rule

Stop early when required columns or valid ranges are absent.

4

Watch out

A successful read is not proof that dates, categories or numbers were parsed correctly.

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Apply: Import and Validate Data

Interpretation

Validation protects every downstream chart and model.

R PATTERN

```
weather <- read.csv("weather.csv",
stringsAsFactors=FALSE)
stopifnot(all(c("Month","Ozone") %in%
names(weather)))
summary(weather)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Export and Audit

1

Meaning

Export writes analysis results in a shareable format while preserving an audit trail.

2

How it works

write.csv() writes tables; saveRDS() preserves one R object; save() preserves multiple objects.

3

Decision rule

Export only the fields needed by the recipient and record the row count and timestamp.

4

Watch out

Writing row names to CSV by default creates an unwanted first column.

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Apply: Export and Audit

Interpretation

An export should be traceable back to its input and script.

R PATTERN

```
write.csv(result, "analysis_output.csv",
row.names=FALSE)
saveRDS(result, "analysis_output.rds")
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Lab 5: Packages and Built-in Datasets

1

Scenario

A junior analyst must document a reproducible package and dataset setup.

2

Output

A setup script and mtcars/quakes data profile.

3

Concepts

packages, namespaces, documentation, str, summary

4

Mapping

LO3 · K3 · A2

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VERIFY THE RESULT

Lab 5: Evidence and Acceptance

1

Evidence 1

Script uses requireNamespace() before optional installation.

2

Evidence 2

Data profile reports correct dimensions.

3

Evidence 3

Package and R versions are recorded.

4

Detailed procedure

Follow the Learner Guide and labs/lab-05-packages-built-in-data/README.pdf. No click-by-click steps are placed in these slides.

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Lab 6: Import, Validate and Export Weather Data

1

Scenario

A public-environment team needs a defensible May ozone summary.

2

Output

A validated weather dataset and audit-ready summary export.

3

Concepts

paths, CSV/Excel import, validation, export

4

Mapping

LO3 · K6 · A2 · A5 · A9

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VERIFY THE RESULT

Lab 6: Evidence and Acceptance

1

Evidence 1

Validation stops the script if a required column is absent.

2

Evidence 2

May average uses only non-missing ozone records.

3

Evidence 3

Export contains source row count and analysis timestamp.

4

Detailed procedure

Follow the Learner Guide and labs/lab-06-import-validate-export/README.pdf. No click-by-click steps are placed in these slides.

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K3 · K6 · A2 · A5 · A9 · CHECKPOINT

Topic 03 Recap

1

Explain

Describe the central ideas in R Packages and Data I/O using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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04

TOPIC 04 · K3 · K5 · A3 · A4 · A7 · A10

Data Visualisation

scatter · line · boxplot · bar · pie · histogram · interpretation

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Chart Selection

1

Meaning

A chart is a visual answer to a specific analytical question, not decoration.

2

How it works

Relationship → scatter; trend → line; distribution → histogram/boxplot; category comparison → bar.

3

Decision rule

Choose the visual encoding that makes the comparison easiest and most accurate.

4

Watch out

Starting with a preferred chart type can hide the structure of the data.

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Apply: Chart Selection

Interpretation

State the question first, then choose the chart.

R PATTERN

```
plot(mpg ~ wt, data=mtcars, pch=19,
col="#1F6FEB")
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Scatter and Line Plots

1

Meaning

Scatter plots reveal association between two numeric variables; lines emphasise ordered progression.

2

How it works

plot(x,y) creates points; lines() or type='l' connects an ordered series.

3

Decision rule

Do not connect unordered observations because the line implies sequence.

4

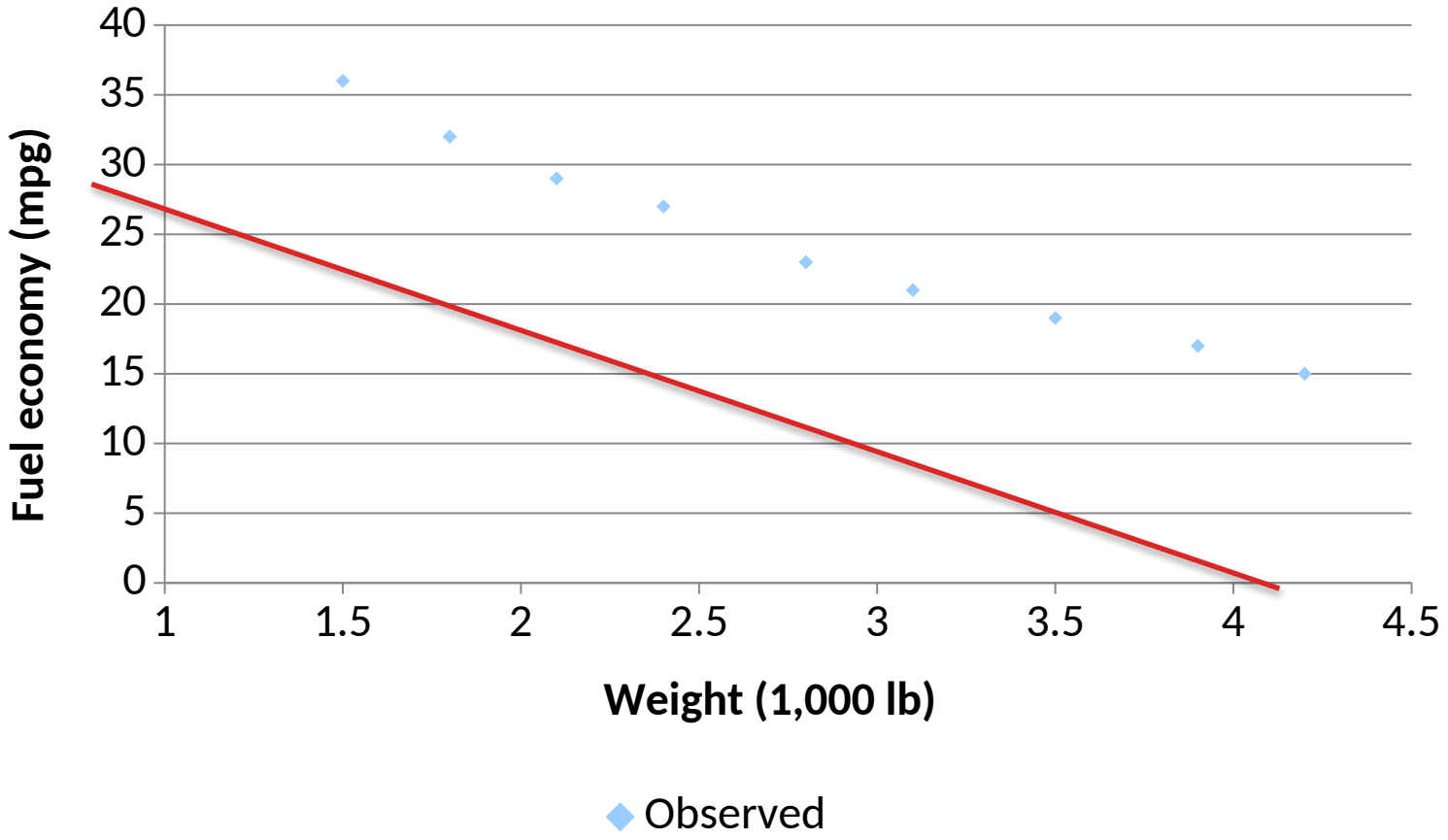
Watch out

Association does not prove causation, even when the point cloud is tight.

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EDITABLE POWERPOINT CHART

Apply: Scatter and Line Plots



READ THE EVIDENCE

Read direction, form, strength and unusual points.

Native chart · editable data

© Tertiary Infotech Academy Pte Ltd

Multiple Scatter Plots

1

Meaning

A scatterplot matrix compares every pair among several numeric variables.

2

How it works

plot(data_frame) creates the matrix; consistent scales and labels support scanning.

3

Decision rule

Use it for exploration, then focus the final story on the most decision-relevant pair.

4

Watch out

A dense matrix becomes unreadable when too many variables are included.

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Apply: Multiple Scatter Plots

Interpretation

Exploration can be broad; communication should be selective.

R PATTERN

```
plot(mtcars[c("mpg","disp","hp","wt")])
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Boxplots

1

Meaning

A boxplot summarises median, quartiles, spread and potential outliers across groups.

2

How it works

The box spans Q1–Q3, the line marks the median and whiskers extend using the 1.5×IQR convention.

3

Decision rule

Use a boxplot to compare group distributions, not merely their means.

4

Watch out

A small group can look deceptively stable; always inspect sample size.

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Apply: Boxplots

Interpretation

Compare centre, spread, overlap and unusual observations.

R PATTERN

```
boxplot(weight ~ feed, data=chickwts,
col="#10B981", las=2)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Bar Charts

1

Meaning

Bar charts compare values or counts across discrete categories from a common zero baseline.

2

How it works

table() creates counts; barplot() maps magnitude to bar length.

3

Decision rule

Sort bars when order is not inherent and keep the baseline at zero.

4

Watch out

Using bars for continuous distributions hides shape and variation.

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Apply: Bar Charts

Interpretation

Bar length is precise; decorative 3D effects are not.

R PATTERN

```
counts <- table(mtcars$am)
barplot(counts,
names.arg=c("Automatic","Manual"))
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Pie Charts

1

Meaning

A pie chart encodes parts of a whole as angles and areas.

2

How it works

Counts are converted to proportions that sum to 100%.

3

Decision rule

Use only for a few categories with a clear whole; prefer bars for close comparisons.

4

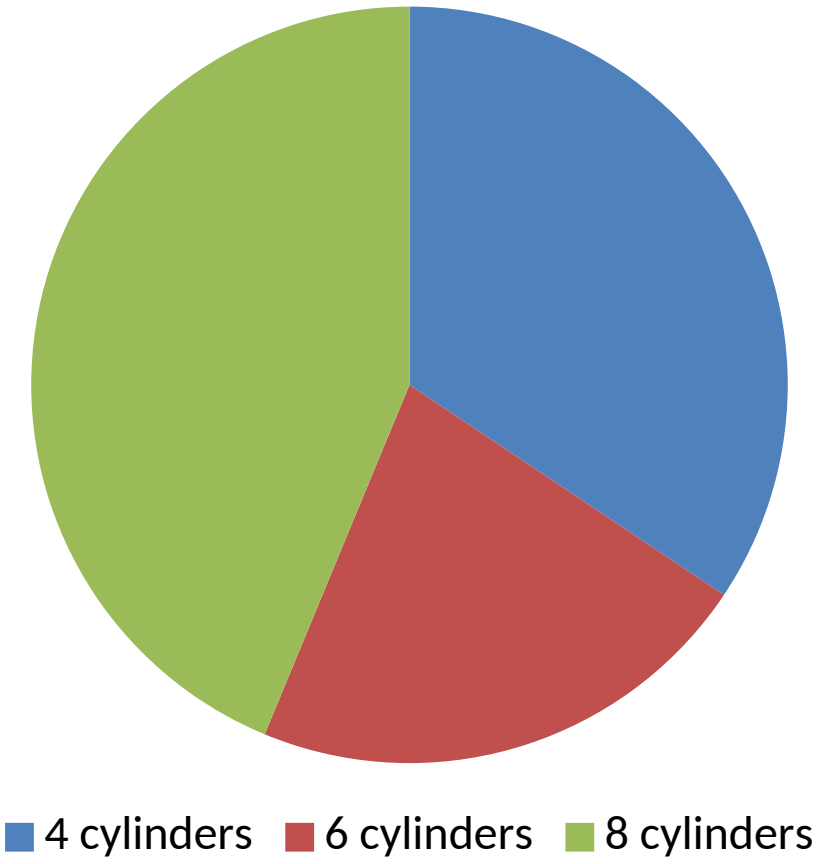
Watch out

Too many slices and similar sizes make ranking difficult.

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EDITABLE POWERPOINT CHART

Apply: Pie Charts



READ THE EVIDENCE

Use a pie only when the part-to-whole message matters.

Native chart · editable data

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TOPIC 04 · K3 · K5 · A3 · A4 · A7 · A10

Histograms

1

Meaning

A histogram groups continuous observations into bins to reveal distribution shape.

2

How it works

Breaks control bin boundaries; frequency or density controls the vertical scale.

3

Decision rule

Try defensible bin widths and report the choice when it affects the story.

4

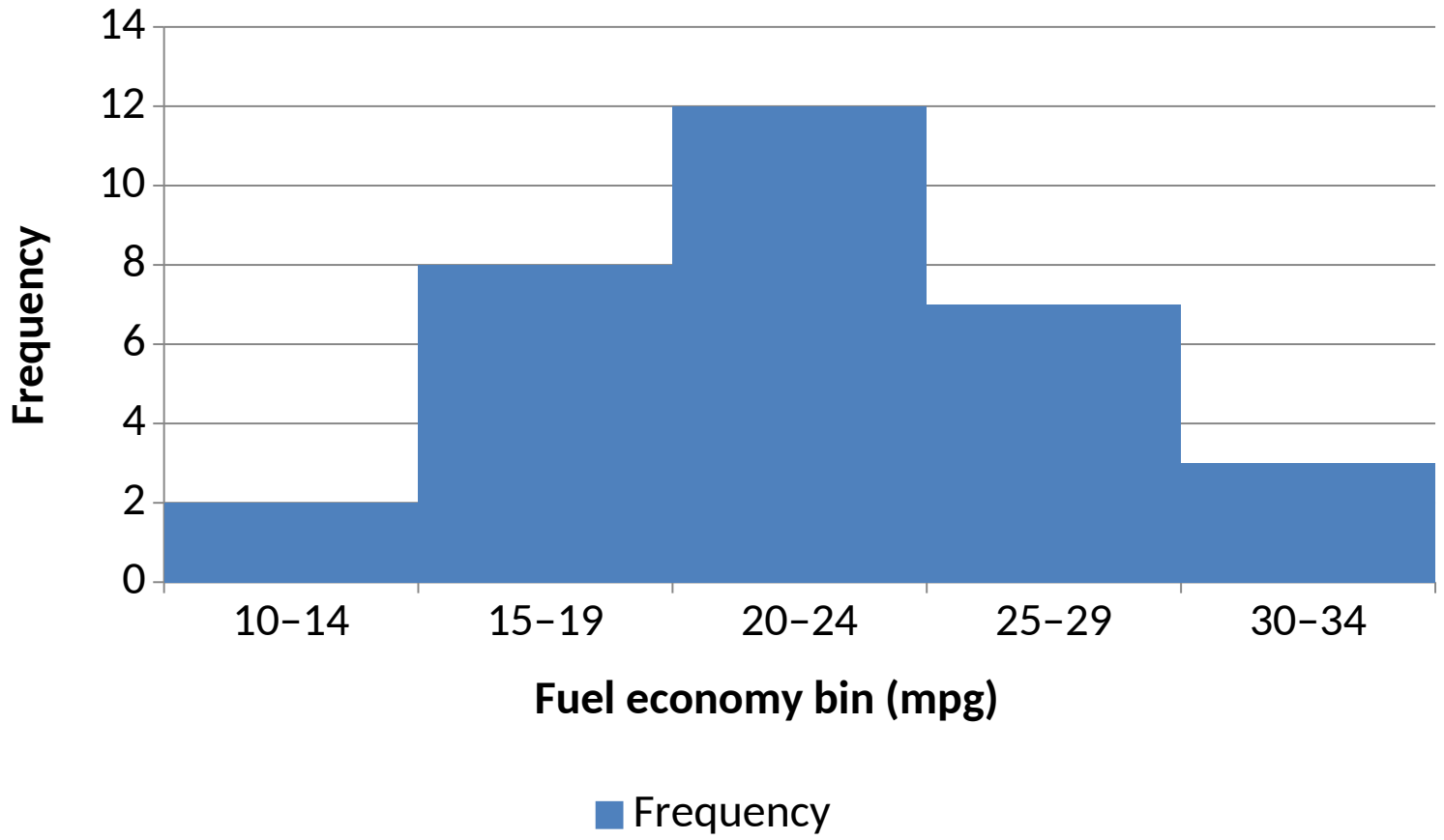
Watch out

A single bin setting can create or hide apparent peaks.

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EDITABLE POWERPOINT CHART

Apply: Histograms



READ THE EVIDENCE

Read centre, spread, skew, modes and unusual gaps.

Native chart · editable data

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Lab 7: Base R Chart Selection

1

Scenario

A transport team needs three views of fuel-economy and transmission data.

2

Output

A scatter plot, bar chart and histogram saved as PNG files.

3

Concepts

scatter, bar, histogram, labels, interpretation

4

Mapping

LO4 · K3 · A3 · A4

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VERIFY THE RESULT

Lab 7: Evidence and Acceptance

1

Evidence 1

Each chart answers a distinct question.

2

Evidence 2

Axes, units and categories are readable.

3

Evidence 3

Interpretations describe evidence without claiming causation.

4

Detailed procedure

Follow the Learner Guide and labs/lab-07-base-r-chart-selection/README.pdf. No click-by-click steps are placed in these slides.

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Lab 8: Compare Distributions with Boxplots

1

Scenario

An agriculture team compares chick weights across feed groups.

2

Output

A grouped boxplot, descriptive table and visual interpretation.

3

Concepts

boxplot, quartiles, outliers, group comparison

4

Mapping

LO4 · K5 · A3 · A4 · A7 · A10

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VERIFY THE RESULT

Lab 8: Evidence and Acceptance

1

Evidence 1

Boxplot includes every feed group.

2

Evidence 2

Summary table agrees with the plotted medians and spread.

3

Evidence 3

Interpretation distinguishes observation from causal claim.

4

Detailed procedure

Follow the Learner Guide and labs/lab-08-distribution-comparison/README.pdf. No click-by-click steps are placed in these slides.

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K3 · K5 · A3 · A4 · A7 · A10 · CHECKPOINT

Topic 04 Recap

1

Explain

Describe the central ideas in Data Visualisation using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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05

TOPIC 05 · K6 · K7 · A7 · A9

R Programming

conditions · loops · next/break · operators · functions · arguments

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Conditions

1

Meaning

if and else choose which block of code runs based on one TRUE/FALSE condition.

2

How it works

Evaluate a scalar condition; run the first matching branch; use ifelse() for vectorised element-wise choices.

3

Decision rule

Make branches mutually understandable and test boundary values.

4

Watch out

Using a vector condition in if() checks only length-one logic and may error.

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Apply: Conditions

Interpretation

A condition converts a rule into an explicit decision.

R PATTERN

```
risk <- 0.72
label <- if (risk >= 0.7) "High" else "Review"
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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While Loops

1

Meaning

A while loop repeats as long as its condition remains TRUE.

2

How it works

Initialise state, test the condition, perform work, then update state.

3

Decision rule

Use when the number of repetitions is not known in advance.

4

Watch out

If state never changes, the loop can run forever.

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Apply: While Loops

Interpretation

Every while loop needs a clear exit path.

R PATTERN

```
x <- 1
while (x <= 5) {
  print(x)
  x <- x + 1
}
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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For Loops

1

Meaning

A for loop processes each element of a known sequence.

2

How it works

The loop variable takes one value at a time; results should normally be preallocated.

3

Decision rule

Use a loop when each iteration has stateful or multi-step logic that is clearer than vectorisation.

4

Watch out

Growing an object repeatedly inside the loop is inefficient.

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Apply: For Loops

Interpretation

Clarity matters more than avoiding loops at all costs.

R PATTERN

```
out <- numeric(5)
for (i in seq_along(out)) {
  out[i] <- i^2
}
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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next and break

1

Meaning

next skips the remainder of the current iteration; break exits the loop completely.

2

How it works

Both alter control flow and should sit beside a clearly stated condition.

3

Decision rule

Use next for invalid records; use break when the goal has been reached or continued work is unsafe.

4

Watch out

An early exit without a message can make missing output look like a successful run.

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Apply: next and break

Interpretation

Explain why records are skipped or processing stops.

R PATTERN

```
for (x in c(3, NA, 7, 12)) {
  if (is.na(x)) next
  if (x > 10) break
  print(x)
}
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Operators

1

Meaning

Arithmetic, comparison and logical operators form expressions that transform data and test rules.

2

How it works

Use + - * / ^ %%, comparisons, and element-wise &, |, !; parentheses make precedence explicit.

3

Decision rule

Vectorised logical operators are normally required for data filtering.

4

Watch out

Floating-point equality can fail due to representation; compare with a tolerance.

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Apply: Operators

Interpretation

Readable expressions reduce logical defects.

R PATTERN

```
keep <- !is.na(mtcars$mpg) & mtcars$mpg >= 20
mtcars[keep, ]
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Functions and Arguments

1

Meaning

A function packages reusable behaviour behind named inputs and a return value.

2

How it works

function(...) defines; default and named arguments make calls flexible; ... forwards optional arguments.

3

Decision rule

Use a function when the same transformation appears twice or needs its own test.

4

Watch out

A missing required argument causes an error; a bad default can hide an assumption.

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Apply: Functions and Arguments

Interpretation

A small, focused function is easier to test than repeated code.

R PATTERN

```
summarise_scores <- function(x, trim=0) {
  mean(x, trim=trim, na.rm=TRUE)
}
summarise_scores(c(1,2,100), trim=.1)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Lab 9: Conditions, Loops, next and break

1

Scenario

A quality team scans daily temperatures and handles invalid records.

2

Output

A controlled processing loop with an exception log.

3

Concepts

if/else, for, while, next, break, logical operators

4

Mapping

LO5 · K6 · A7 · A9

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VERIFY THE RESULT

Lab 9: Evidence and Acceptance

1

Evidence 1

Missing values are skipped and counted.

2

Evidence 2

Safety stop is explicit and testable.

3

Evidence 3

Loop and vectorised count agree before the stop point.

4

Detailed procedure

Follow the Learner Guide and labs/lab-09-conditions-loops/README.pdf. No click-by-click steps are placed in these slides.

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Lab 10: Build and Test Reusable Functions

1

Scenario

A game-based learning team needs a reliable two-dice simulator and summary function.

2

Output

Two documented functions with deterministic tests.

3

Concepts

functions, default arguments, named arguments, ..., set.seed

4

Mapping

LO5 · K7 · A7 · A9

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VERIFY THE RESULT

Lab 10: Evidence and Acceptance

1

Evidence 1

Invalid n or sides values stop with clear messages.

2

Evidence 2

Default and named argument calls both work.

3

Evidence 3

Tests are reproducible under the recorded seed.

4

Detailed procedure

Follow the Learner Guide and labs/lab-10-reusable-functions/README.pdf. No click-by-click steps are placed in these slides.

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Topic 05 Recap

1

Explain

Describe the central ideas in R Programming using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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06

TOPIC 06 · K1 · K2 · K5 · A1 · A3 · A4 · A6 · A7 · A8

Statistical Analysis with R

descriptive statistics · correlation · regression · tests · ANOVA · diagnostics

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Descriptive Statistics

1

Meaning

Descriptive statistics summarise observed data without generalising beyond it.

2

How it works

Location: mean/median; spread: range/IQR/SD; shape: skew and unusual values; counts: n and missingness.

3

Decision rule

Report centre, spread and valid sample size together.

4

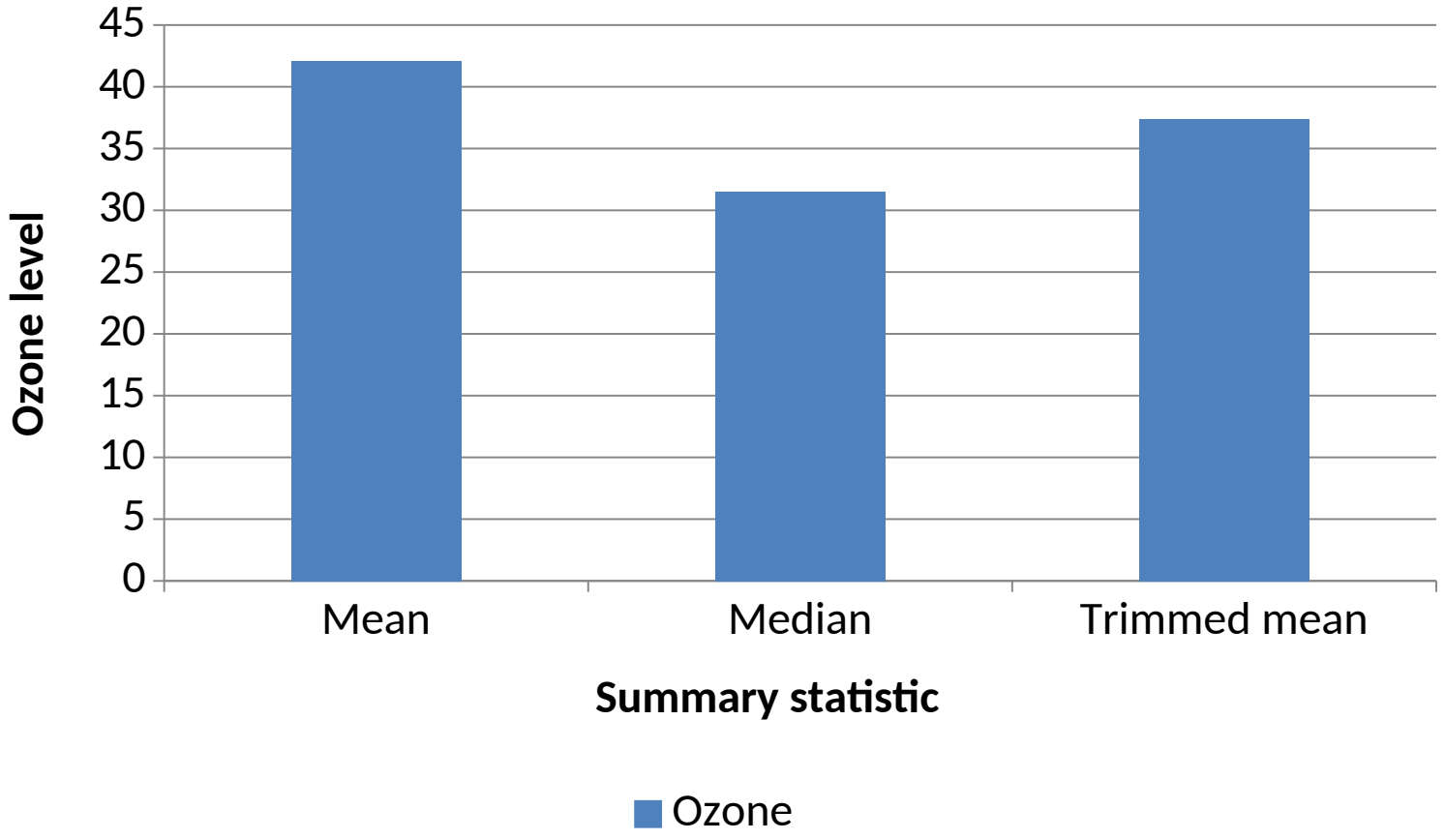
Watch out

Averages without context can conceal missing data or a highly skewed distribution.

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EDITABLE POWERPOINT CHART

Apply: Descriptive Statistics



READ THE EVIDENCE

Describe the data before modelling it.

Native chart · editable data

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Trimmed Mean and Robustness

1

Meaning

A trimmed mean removes equal proportions from both tails before averaging.

2

How it works

Increasing trim reduces the influence of extreme values while still using more information than the median.

3

Decision rule

Use a robust summary when outliers are plausible but should not dominate the typical value.

4

Watch out

Trimming should be justified and cannot replace investigating data quality.

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Apply: Trimmed Mean and Robustness

Interpretation

Robustness is a deliberate trade-off between sensitivity and stability.

R PATTERN

```
ozone <- airquality$Ozone
c(mean=mean(ozone,na.rm=TRUE),
trim10=mean(ozone,trim=.1,na.rm=TRUE),
median=median(ozone,na.rm=TRUE))
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Correlation

1

Meaning

Correlation measures the strength and direction of a linear relationship between two numeric variables.

2

How it works

Pearson r ranges from -1 to +1; sign gives direction and magnitude gives linear strength.

3

Decision rule

Inspect a scatter plot and outliers before interpreting r.

4

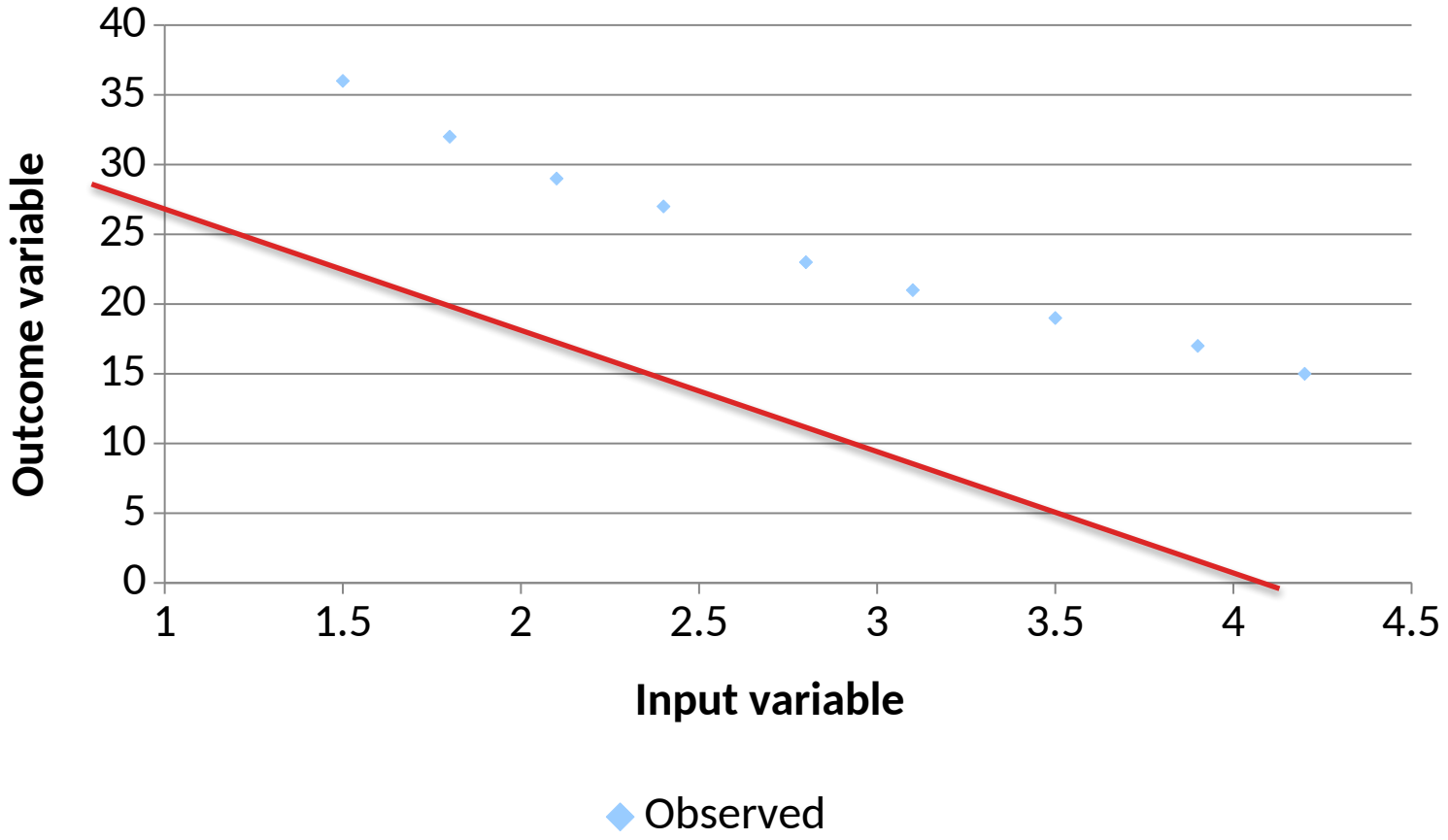
Watch out

Correlation does not establish causation and may miss non-linear relationships.

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EDITABLE POWERPOINT CHART

Apply: Correlation



READ THE EVIDENCE

A negative r means one variable tends to decrease as the other increases.

Native chart · editable data

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Linear Regression

1

Meaning

Simple linear regression estimates a line that predicts an outcome from one input.

2

How it works

lm(y ~ x) estimates intercept and slope by minimising squared residuals.

3

Decision rule

Use when the relationship is plausibly linear and diagnostics support the assumptions.

4

Watch out

Extrapolating beyond the observed x range can produce implausible predictions.

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Apply: Linear Regression

Interpretation

The slope quantifies the expected change in y for one unit of x.

R PATTERN

```
model <- lm(mpg ~ wt, data=mtcars)
coef(model)
predict(model, data.frame(wt=3))
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Residuals and Mean Squared Error

1

Meaning

A residual is observed minus predicted; MSE averages squared prediction errors.

2

How it works

Squaring makes errors positive and penalises large misses more strongly.

3

Decision rule

Inspect residual plots as well as one error metric.

4

Watch out

A low training MSE does not guarantee accurate new predictions.

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Apply: Residuals and Mean Squared Error

Interpretation

Residual patterns reveal model problems that one score can hide.

R PATTERN

```
model <- lm(mpg ~ wt, mtcars)
res <- residuals(model)
mean(res^2)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Regression Diagnostics

1

Meaning

Diagnostics test whether the linear model is an adequate representation of the data.

2

How it works

Check linearity, constant variance, normality of residuals and influential observations.

3

Decision rule

Change the model or interpretation when assumptions fail materially.

4

Watch out

A significant coefficient is not proof that the full model is useful.

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Apply: Regression Diagnostics

Interpretation

Diagnose unintended outcomes before trusting a model.

R PATTERN

```
model <- lm(mpg ~ wt, mtcars)
par(mfrow=c(2,2))
plot(model)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Multiple Regression

1

Meaning

Multiple regression predicts one outcome from several inputs and can include interactions.

2

How it works

Each coefficient is conditional on the other variables; x1:x2 represents an interaction.

3

Decision rule

Choose predictors from the question and domain logic, not only automated significance.

4

Watch out

Highly correlated predictors make coefficients unstable and difficult to interpret.

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Apply: Multiple Regression

Interpretation

More predictors do not automatically mean a better model.

R PATTERN

```
model <- lm(mpg ~ wt + hp + disp + wt:hp,
mtcars)
summary(model)
predict(model,
data.frame(wt=2.6,hp=110,disp=160))
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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R-squared and Model Fit

1

Meaning

R-squared is the proportion of observed outcome variance explained by the fitted model in the sample.

2

How it works

Adjusted R-squared penalises unnecessary predictors; both must be read with residual diagnostics.

3

Decision rule

Judge usefulness against the decision context, not a universal threshold.

4

Watch out

A high R-squared can coexist with bias, leakage or poor out-of-sample performance.

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Apply: R-squared and Model Fit

Interpretation

Model fit is evidence, not a guarantee of accuracy.

R PATTERN

```
model <- lm(mpg ~ wt + hp, mtcars)
summary(model)$r.squared
summary(model)$adj.r.squared
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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Hypothesis Testing

1

Meaning

A hypothesis test evaluates how compatible the observed data are with a stated null hypothesis.

2

How it works

Define H0/H1, choose alpha and test, compute a statistic and p-value, then decide and interpret.

3

Decision rule

Use 'reject' or 'fail to reject' H0; do not claim that a non-significant result proves equality.

4

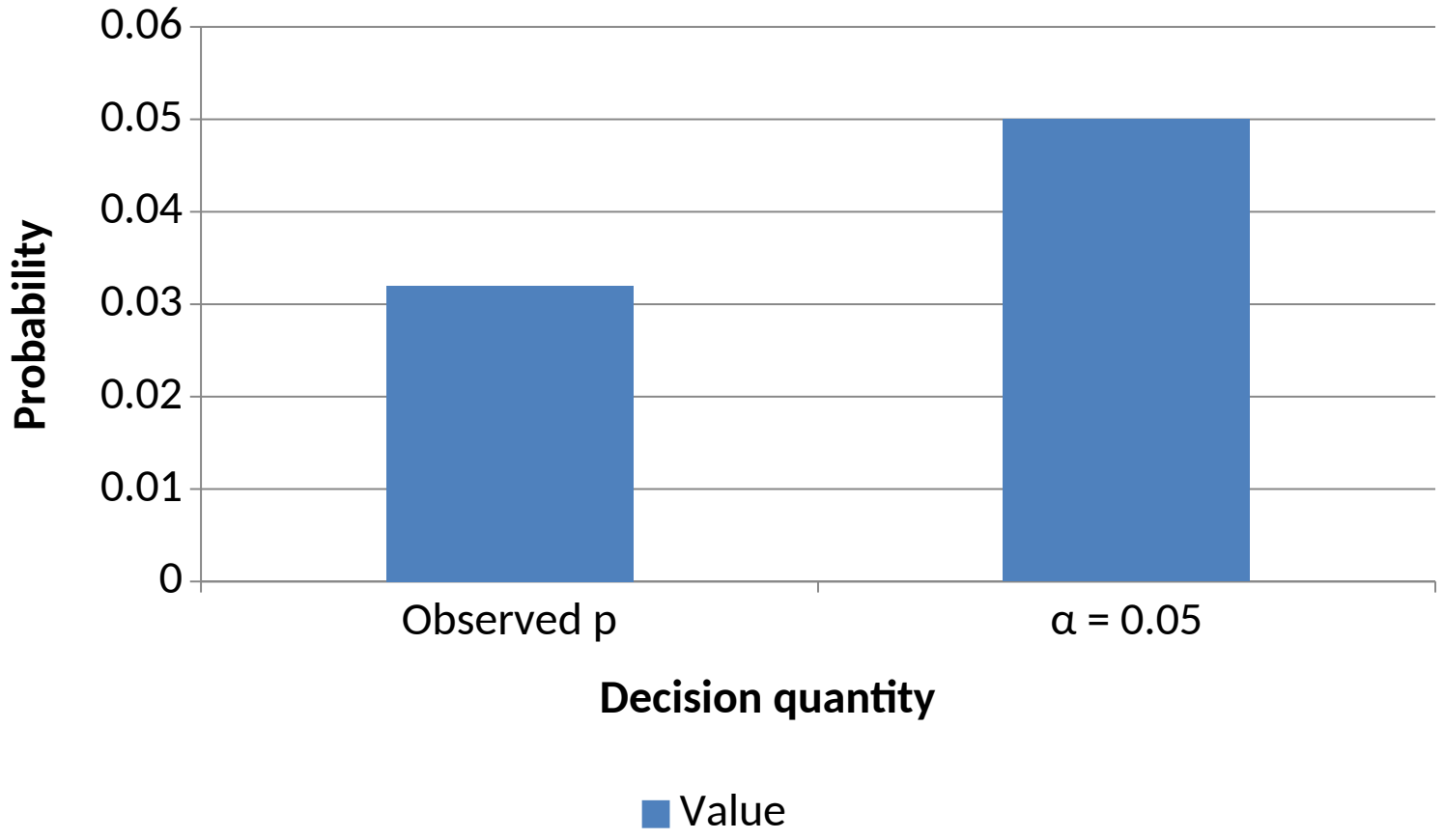
Watch out

Statistical significance is not the same as practical importance.

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EDITABLE POWERPOINT CHART

Apply: Hypothesis Testing



READ THE EVIDENCE

Predefine the question and threshold before looking at the result.

Native chart · editable data

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Two-Sample t-Test

1

Meaning

A two-sample t-test compares the means of two independent groups.

2

How it works

Welch's test is the R default and does not assume equal variances; alternative controls two- or one-sided questions.

3

Decision rule

Use a two-sided test unless a directional hypothesis was defined in advance.

4

Watch out

Running both directions after seeing the data inflates false-positive risk.

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Apply: Two-Sample t-Test

Interpretation

Report the group estimates, confidence interval and p-value.

R PATTERN

```
t.test(weight ~ group, data=sleep,
alternative="two.sided")
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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One-Sample t-Test

1

Meaning

A one-sample t-test evaluates whether a sample mean differs from a reference value.

2

How it works

t.test(x, mu=reference) compares the observed standardised difference with the t distribution.

3

Decision rule

Use when observations are independent and the distribution is reasonably compatible with the test.

4

Watch out

A large sample can make a tiny, unimportant difference statistically significant.

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Apply: One-Sample t-Test

Interpretation

Pair the p-value with the estimated difference and interval.

R PATTERN

```
meatmeal <- subset(chickwts, feed=="meatmeal")
$weight
t.test(meatmeal, mu=160)
```

Run this pattern with the lab's own data and validate the result before interpreting it.

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ANOVA

1

Meaning

One-way ANOVA tests whether at least one group mean differs among three or more independent groups.

2

How it works

It compares between-group variation with within-group variation using an F statistic.

3

Decision rule

If significant, follow with planned contrasts or a multiple-comparison method to locate differences.

4

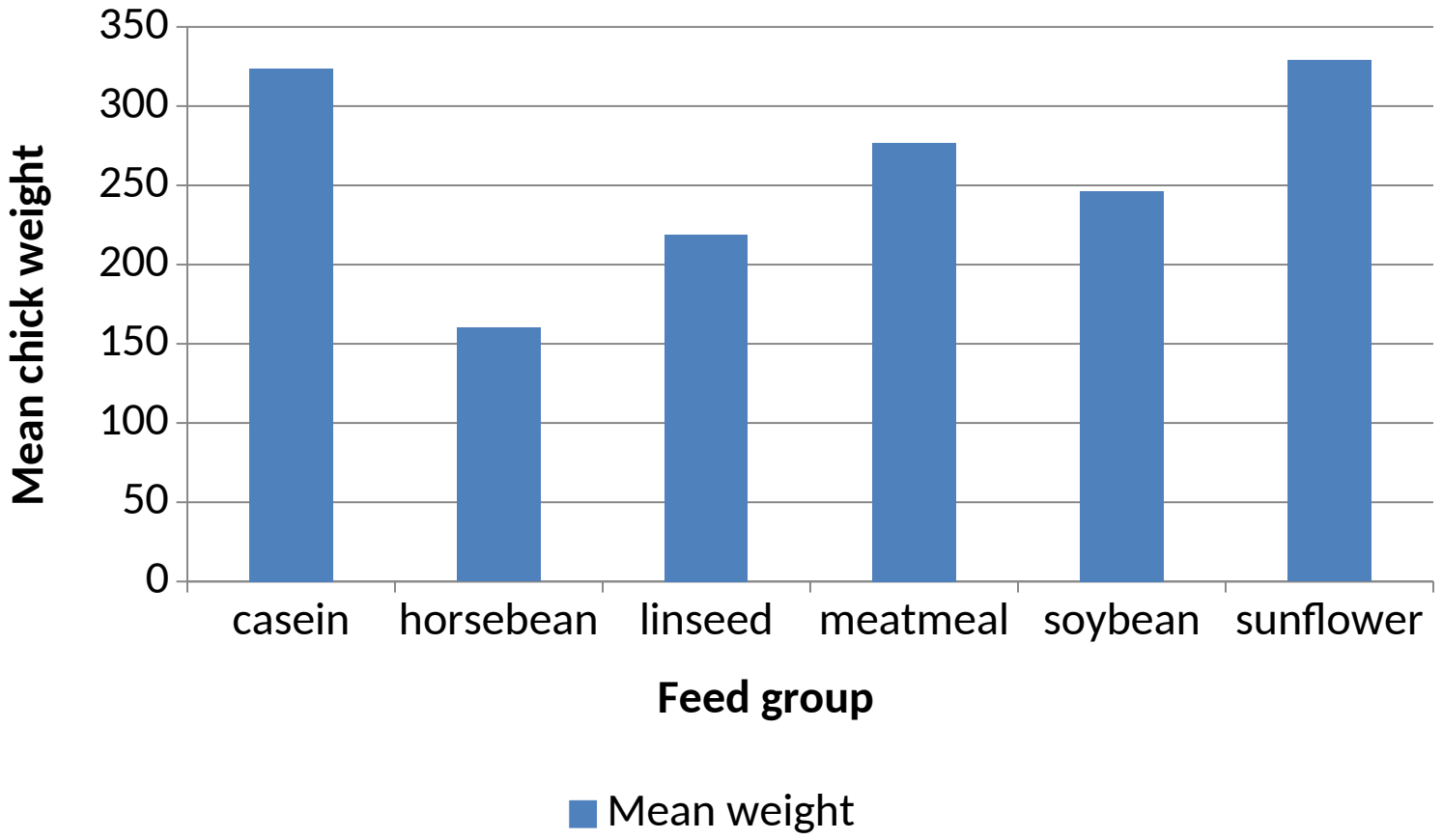
Watch out

A significant ANOVA does not say every group differs from every other group.

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EDITABLE POWERPOINT CHART

Apply: ANOVA



READ THE EVIDENCE

ANOVA answers an overall question; post-hoc analysis answers which groups differ.

Native chart · editable data

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ADVANCED · HANDS-ON OVERVIEW

Lab 11: Descriptive Statistics, Correlation and Regression

1

Scenario

A health analytics team studies height, weight and a numeric outcome.

2

Output

A validated summary, correlation result, fitted model and diagnostic evidence.

3

Concepts

descriptive statistics, correlation, lm, residuals, R-squared

4

Mapping

LO6 · K1 · K2 · A1 · A3 · A4 · A6

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VERIFY THE RESULT

Lab 11: Evidence and Acceptance

1

Evidence 1

Correlation interpretation includes direction and strength.

2

Evidence 2

Model formula and prediction target are explicit.

3

Evidence 3

Diagnostic evidence is included before a conclusion.

4

Detailed procedure

Follow the Learner Guide and labs/lab-11-descriptive-correlation-regression/README.pdf. No click-by-click steps are placed in these slides.

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ADVANCED · HANDS-ON OVERVIEW

Lab 12: Hypothesis Testing and ANOVA Capstone

1

Scenario

A health-policy team compares groups and proposes a defensible model update.

2

Output

A t-test, ANOVA, post-hoc comparison and revised analysis decision.

3

Concepts

H0/H1, p-value, t-test, ANOVA, diagnostics, remediation

4

Mapping

LO6 · K5 · A1–A10

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VERIFY THE RESULT

Lab 12: Evidence and Acceptance

1

Evidence 1

Every conclusion uses reject/fail-to-reject language correctly.

2

Evidence 2

ANOVA conclusion does not overstate which groups differ.

3

Evidence 3

The updated script reruns successfully and the audit note explains the change.

4

Detailed procedure

Follow the Learner Guide and labs/lab-12-hypothesis-anova-capstone/README.pdf. No click-by-click steps are placed in these slides.

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K1 · K2 · K5 · A1 · A3 · A4 · A6 · A7 · A8 · CHECKPOINT

Topic 06 Recap

1

Explain

Describe the central ideas in Statistical Analysis with R using precise R/statistical language.

2

Apply

Choose the relevant R structure, function, chart or statistical method for the question.

3

Validate

Check object structure, missingness, assumptions and output before interpreting.

4

Evidence

Save the script, workbook and verification result in the matching lab folder.

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COURSE SUMMARY

What You Achieved

1

LO1

Install R and RStudio IDE and work confidently with scripts, the console and help system.

2

LO2

Select, create, inspect and manipulate R data types and data structures.

3

LO3

Use R packages and datasets and import, validate and export external data.

4

LO4

Create and interpret appropriate data visualisations in R.

5

LO5

Write R programs using conditions, loops and reusable functions.

6

LO6

Apply descriptive statistics, correlation, regression, hypothesis testing and ANOVA in R.

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Courseware and Assessment on the LMS

DOWNLOAD COURSEWARE

Open the LMS course page and download the learner slides, Learner Guide and lab materials before the assessment.

COMPLETE ASSESSMENT

Take assessment attendance, complete the WA and Case Study, then upload both answer documents and sign the summary record.

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FINAL CHECK

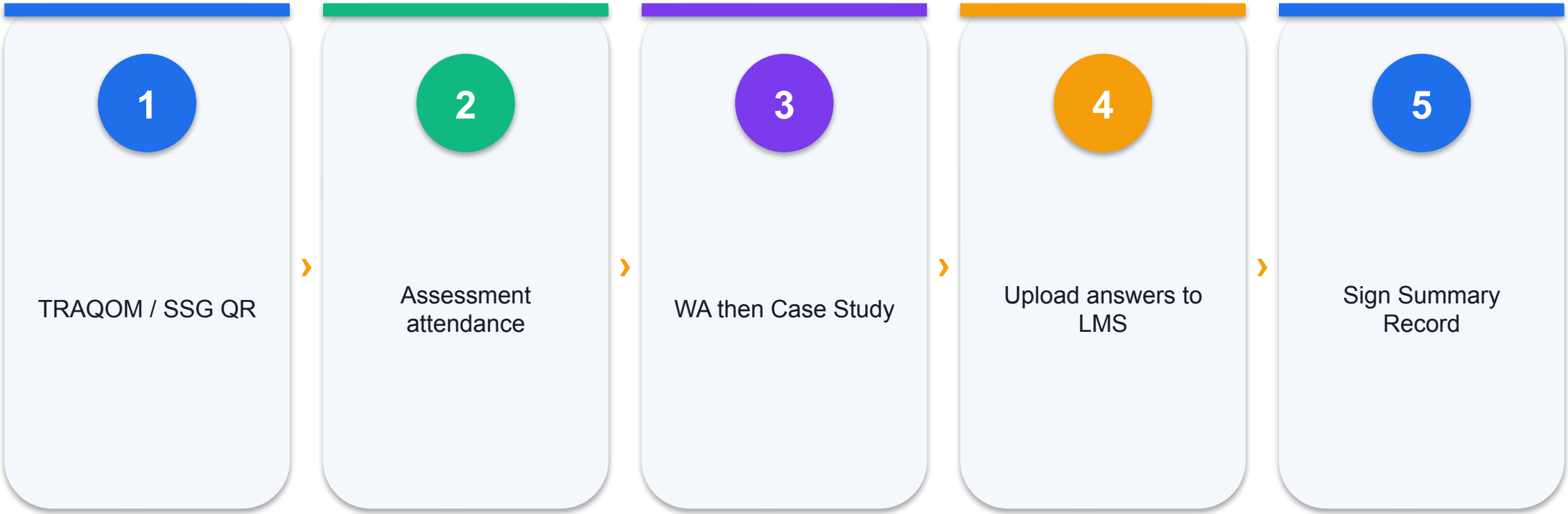
Assessment Reminder

1. 60-minute WA
2. 80-minute Case Study
3. Open book
4. Upload on LMS
5. Sign record

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ASSESSMENT DAY

Assessment Flow



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Digital Attendance (Mandatory)

1

Assessment QR

Scan the assessment digital attendance QR before starting.

2

Identity

Use your own Singpass/mobile device and verify the submitted record.

3

Completion

Remain until both instruments are submitted and the record is signed.

4

Support

Tell the trainer immediately if the QR or submission does not work.

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Thank You

Build the script. Check the evidence. Explain the decision.

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